NEXT ENGINEER · _SHEERAT_

Virology

BOE088

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 8 | **Credits:** 3 | **L-T-P:** 3-0-0

Scheme: AKTU B.Tech 4th year, Open Elective-III, 2025-26 onwards

UNIT 1 Introduction to Viruses

UNIT 2 Virus-Host Interactions and Diagnosis

UNIT 3 Classification, Positive-Strand RNA and DNA Viruses

UNIT 4 Negative-Strand RNA, dsRNA Viruses and Retroviruses

UNIT 5 Vaccines, Antivirals and RNA-Based Therapies

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: the biology, with comparison tables and labelled-diagram descriptions.
3. **Solved examples**: short calculations examiners like (plaque assay titre, multiplicity of infection, RT-PCR Ct values, test predictive values, herd immunity threshold, CD4 counts).
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (history, structure, genomes, isolation and purification) and Unit 2 (effects on host, interferons, viral evasion, diagnostics).
- **Afternoon (3 hours):** Unit 3 (Baltimore classification, picorna, flavi, corona, small and large DNA viruses) and Unit 4 (paramyxo, influenza, rabies, rotavirus, HIV).
- **Evening (2 hours):** Unit 5 (vaccines, adjuvants, manufacturing QC, antivirals, interferons, antisense, siRNA, ribozymes), then the "Last-hour checklist".
- **In the exam:** draw a labelled virus structure (capsid, envelope, spikes, genome) and the replication cycle for the virus asked. Name one disease and one drug or vaccine for each family.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Introduction to Viruses

✍️ 2-mark definitions

- **Virus:** an obligate intracellular parasite made of a nucleic acid genome (DNA or RNA, never both) enclosed in a protein coat, which replicates only inside living host cells using the host's machinery.
- **Virion:** a complete, infectious virus particle outside the cell.
- **Capsid:** the protein shell around the viral genome, made of repeating subunits called **capsomeres**.
- **Nucleocapsid:** the capsid together with the enclosed genome.
- **Envelope:** a lipid bilayer derived from the host cell membrane surrounding some virions, carrying viral glycoprotein spikes (peplomers).
- **Bacteriophage:** a virus that infects bacteria (e.g. T4, lambda).
- **Viroid:** a small, circular, naked infectious RNA (about 250–400 nucleotides) without a protein coat, causing plant diseases (potato spindle tuber viroid).
- **Prion:** an infectious misfolded protein (PrP^{sc}) without nucleic acid that causes transmissible spongiform encephalopathies (CJD, mad cow disease); not a virus.
- **Cytopathic effect (CPE):** visible changes in cultured cells caused by viral infection (rounding, lysis, syncytia, inclusion bodies).
- **Plaque:** a clear zone in a cell monolayer or bacterial lawn formed by lysis of cells by one infectious virus and its progeny; the basis of plaque assays (PFU).

💡 Core concepts

History of virology

Year	Milestone
1796	Edward Jenner uses cowpox to protect against smallpox (first vaccine)
1885	Louis Pasteur's rabies vaccine
1892	Dmitri Ivanovsky shows tobacco mosaic disease agent passes bacteria-proof filters
1898	Martinus Beijerinck calls it "contagium vivum fluidum" (living infectious fluid); Loeffler and Frosch find foot-and-mouth disease virus (first animal virus)
1901	Walter Reed: yellow fever virus (first human virus) spread by mosquitoes
1915–17	Twort and d'Herelle discover bacteriophages
1935	Wendell Stanley crystallizes TMV
1939	First electron micrograph of TMV
1949	Enders, Weller, Robbins grow poliovirus in cell culture
1950s	Salk (inactivated) and Sabin (oral) polio vaccines

Year	Milestone
1970	Temin and Baltimore discover reverse transcriptase
1980	WHO declares smallpox eradicated
1983	HIV identified (Montagnier, Barré-Sinoussi)
2003, 2012, 2019	SARS-CoV, MERS-CoV, SARS-CoV-2 emerge
2014	India declared polio-free (with the WHO South-East Asia Region)
2020	First mRNA vaccines authorized

Properties of viruses: acellular; contain only one type of nucleic acid; no ribosomes or energy-generating metabolism; replicate by assembly from components, not by binary fission; filterable (pass bacterial filters); not affected by antibiotics; can be crystallized.

Diversity: viruses infect all life forms: animals, plants, fungi, bacteria (phages), archaea, even other viruses (virophages). They are the most abundant biological entities (about 10^{31} phages on Earth).

Shapes (symmetry)

Symmetry	Description	Examples
Helical	Capsomeres arranged in a helix around the genome; rod or filamentous	TMV, influenza and rabies nucleocapsids, Ebola
Icosahedral	20 triangular faces, 12 vertices; efficient closed shell	Adenovirus, poliovirus, herpesvirus capsid
Complex	Neither purely helical nor icosahedral	Bacteriophage T4 (head, tail, tail fibres), poxviruses
Enveloped	Lipid envelope over helical or icosahedral nucleocapsid; pleomorphic	Influenza, HIV, SARS-CoV-2, herpes

Sizes: 17–30 nm (circoviruses, parvoviruses ~20 nm, poliovirus ~30 nm) up to ~200–400 nm (poxviruses, visible by light microscopy at the limit); giant viruses (Mimivirus ~750 nm, Pandoravirus ~1 µm). Measured by electron microscopy, ultrafiltration and ultracentrifugation.

Genome components: DNA or RNA; single- or double-stranded; linear or circular; segmented or non-segmented; positive (mRNA-sense) or negative sense; size from ~1.7 kb (circovirus) to over 2 Mb (Pandoravirus). Genomes carry genes for structural proteins, replication enzymes (polymerases), regulatory and host-modulating proteins; special features: terminal repeats, IRES, cap, poly-A, packaging signals.

Virus structure (draw it): genome → capsid (capsomeres) → nucleocapsid → matrix protein (in some enveloped viruses) → envelope with glycoprotein spikes. Some virions carry enzymes (RNA polymerase in negative-strand viruses; reverse transcriptase in retroviruses).

General replication cycle (6 steps): (1) attachment (adsorption) to receptors, (2) penetration/entry (endocytosis, membrane fusion, injection), (3) uncoating, (4) synthesis of mRNA, proteins and genome (biosynthesis), (5) assembly (maturation), (6) release (lysis or budding).

Isolation of viruses

- **Cell (tissue) culture:** primary cells, diploid cell lines (WI-38, MRC-5), continuous cell lines (Vero, HeLa, MDCK); detect growth by CPE, haemadsorption, immunofluorescence.
- **Embryonated chicken eggs:** inoculation of chorioallantoic membrane (pocks, poxviruses), amniotic and allantoic cavities (influenza, used for vaccine production), yolk sac.
- **Laboratory animals:** suckling mice (arboviruses, coxsackie), when culture is not possible.
- **Bacteriophages:** isolated on bacterial lawns.

Purification of viruses: clarification by low-speed centrifugation → concentration by ultracentrifugation, PEG precipitation or ultrafiltration → density-gradient centrifugation (sucrose rate-zonal; CsCl isopycnic) → chromatography (ion exchange, size exclusion, affinity) → check purity (EM, SDS-PAGE, infectivity/particle ratio).

Quantification: plaque assay (PFU/mL), $TCID_{50}$ (50% tissue culture infective dose), haemagglutination titre, qPCR (genome copies), electron microscopy particle counts.

🧮 Solved examples

Example 1: Plaque assay titre. A virus stock is diluted 10^{-6} and 0.1 mL is plated; 45 plaques appear. Titre = plaques / (dilution × volume) = 45 / (10^{-6} × 0.1) = **4.5×10^{8} PFU/mL**.

Example 2: Multiplicity of infection (MOI). 2 × 10^{6} cells are infected with 10^{7} PFU. MOI = 10^{7} / (2 × 10^{6}) = **5**. By the Poisson distribution, the fraction of cells infected = 1 − e^(−MOI) = 1 − e^(−5) ≈ **99.3%**. At MOI = 1, only 1 − e^(−1) = 63.2% of cells are infected.

Example 3: Dilution to plate. You want about 50 plaques per plate (0.1 mL plated) from a 5 × 10^{7} PFU/mL stock. Needed concentration = 50/0.1 = 500 PFU/mL; dilution = 500 / (5 × 10^{7}) = **10^{-5}**.

Key points to remember

- Ivanovsky (1892) and Beijerinck (1898) discovered viruses with TMV; Stanley crystallized TMV (1935).
- Symmetry: helical, icosahedral, complex; enveloped or naked.
- Genome: DNA or RNA, ss or ds, linear or circular, segmented, + or − sense.
- Replication: attachment → entry → uncoating → biosynthesis → assembly → release.
- Isolation: cell culture, eggs, animals; purification: centrifugation, density gradient; titre in PFU/mL.

⭐ Important questions

1. Trace the history of virology with major milestones.
2. Describe the general properties, shapes and sizes of viruses with a labelled diagram.
3. Explain the components of viral genomes and the diversity of viruses.
4. Explain the methods used for isolation of viruses.
5. Explain the purification and quantification of viruses (plaque assay, $TCID_{50}$).

NEXT ENGINEER sheerat

UNIT 2 Virus-Host Interactions and Diagnosis

✍️ 2-mark definitions

- **Lytic infection:** infection ending in host cell lysis and release of progeny virions.
- **Persistent infection:** long-term infection in which the virus continues to be produced at low levels without killing the host cell quickly (hepatitis B).
- **Latent infection:** the viral genome remains in the cell without producing virions, and can reactivate later (herpes simplex in neurons, varicella-zoster causing shingles).
- **Transformation:** conversion of a normal cell into a cancer-like cell by a virus (oncogenic viruses: HPV, HBV, HCV, EBV, HTLV-1, KSHV).
- **Interferons (IFNs):** cytokines produced by infected cells that induce an antiviral state in neighbouring cells (type I: IFN-α, IFN-β; type II: IFN-γ; type III: IFN-λ).
- **Antiviral state:** the condition of a cell exposed to interferon in which hundreds of interferon-stimulated genes (ISGs) block virus replication.
- **Pattern recognition receptors (PRRs):** host sensors that detect viral molecules (TLR3, TLR7/8, TLR9, RIG-I, MDA5, cGAS-STING).
- **RT-PCR:** reverse transcription followed by polymerase chain reaction; detects viral RNA; real-time RT-PCR reports a cycle threshold (Ct).
- **Rapid antigen test (RAT):** a lateral-flow immunoassay that detects viral proteins in a sample in 15–30 minutes.
- **Ct value:** the PCR cycle at which fluorescence crosses the threshold; a lower Ct means more viral nucleic acid.

💡 Core concepts

Effects of virus infection on host cells

- **Cytopathic effects:** cell rounding and detachment, lysis, syncytia (cell fusion: RSV, measles, HIV), inclusion bodies (Negri bodies in rabies, owl's eye in CMV, Guarnieri bodies in poxviruses), vacuolation.
- **Biochemical effects:** shutdown of host protein synthesis (poliovirus cleaves eIF4G), host DNA/RNA synthesis inhibition, cytoskeleton changes, membrane changes.
- **Apoptosis** (programmed cell death) or its inhibition.
- **Transformation** and cancer (HPV E6/E7 inactivate p53 and Rb).
- **Effects on the organism:** inapparent (subclinical), acute, persistent, latent, slow (subacute sclerosing panencephalitis after measles) infections; immunopathology.

Innate antiviral defence: interferon pathway

1. **Detection:** viral RNA/DNA sensed by TLRs (in endosomes) and RIG-I/MDA5 (cytoplasmic RNA) or cGAS-STING (cytoplasmic DNA).
2. **Signalling:** adaptors (MAVS, TRIF, MyD88) activate kinases → transcription factors IRF3, IRF7, NF-κB.
3. **IFN production:** IFN-β and IFN-α secreted.

4. **IFN signalling:** IFN binds IFNAR on neighbouring cells → JAK1/TYK2 → STAT1/STAT2 + IRF9 (ISGF3) → ISRE in DNA → ISGs.
5. **ISG effectors:** PKR (phosphorylates eIF2α, stops translation), OAS/RNase L (degrades RNA), Mx proteins (block influenza), IFITMs (block entry), tetherin (blocks release), APOBEC3 (hypermutates HIV), ISG15. - Also NK cells, complement, inflammation; then adaptive immunity (antibodies, cytotoxic T cells).

Viral counter-mechanisms (immune evasion)

Strategy	Example
Block IFN induction	Influenza NS1 binds RNA and RIG-I/TRIM25; HCV NS3/4A protease cleaves MAVS
Block IFN signalling	Paramyxovirus V proteins degrade STATs
Block ISG action	Adenovirus VA RNA inhibits PKR; vaccinia E3L binds dsRNA
Hide from T cells	HIV Nef and herpes proteins down-regulate MHC-I
Antigenic variation	Influenza drift and shift; HIV high mutation rate
Latency	Herpesviruses hide in neurons or B cells
Decoy receptors, cytokine mimics	Poxvirus soluble IFN receptors; EBV IL-10 homologue
Inhibit apoptosis	Viral Bcl-2 homologues
Infect immune cells	HIV destroys $CD4^+$ T cells

Diagnostics

Method	Detects	Pros	Cons
Real-time RT-PCR	Viral RNA	Gold standard; very sensitive and specific; quantitative (Ct)	Lab, trained staff, 2–6 h, costly
Rapid antigen test	Viral protein	Fast (15–30 min), cheap, point of care	Less sensitive, misses low viral loads
ELISA / serology	Antibodies IgM (recent) and IgG (past)	Exposure history, vaccine response	Window period; cannot detect early infection
Virus isolation in culture	Live virus	Proves infectivity	Slow (days), biosafety
Electron microscopy	Particles by morphology	Rapid identification of unknowns	Insensitive, costly
LAMP, CRISPR-based tests	Nucleic acid (isothermal)	Faster, simpler than PCR	Newer, validation
Next-generation sequencing	Whole genomes	Variants, new viruses	Costly, bioinformatics

RT-PCR steps: sample (nasopharyngeal swab in viral transport medium) → RNA extraction → reverse transcription (cDNA) → PCR amplification with primers and fluorescent probes (TaqMan) for viral genes

(e.g. SARS-CoV-2 E, N, RdRp/ORF1ab) and a human internal control (RNase P) → Ct read-out (e.g. Ct ≤ 35 positive). Each cycle doubles the target, so a difference of 3.3 cycles ≈ 10-fold in template.

Rapid antigen test (lateral flow): sample → conjugate pad (gold-labelled antibody binds antigen) → nitrocellulose strip with test line (capture antibody) and control line → coloured lines appear.

🧮 Solved examples

Example 1: Ct difference. Patient A has Ct 25 and patient B Ct 30 for the same gene. A has 2^(30 − 25) = 2^5 = **32 times** more viral RNA.

Example 2: Predictive values of a rapid antigen test. Sensitivity 80%, specificity 98%, prevalence 5%, 1,000 people tested.

- Infected = 50: true positives = 40, false negatives = 10.
- Not infected = 950: true negatives = 931, false positives = 19.
- **PPV = 40/(40 + 19) = 67.8%**; **NPV = 931/(931 + 10) = 98.9%**.
- So a positive RAT in a low-prevalence setting should be confirmed by RT-PCR.

Example 3: Ten-fold rule. If a standard curve has a slope of −3.32 cycles per $\log_{10}$ (100% efficiency), a sample with Ct 6.64 cycles lower than another has 10^2 = **100 times** more template.

Key points to remember

- Infection types: lytic, persistent, latent, transforming.
- IFN pathway: PRR → MAVS/TRIF → IRF3/7 → IFN-β → IFNAR → JAK-STAT (ISGF3) → ISGs (PKR, OAS, Mx).
- Evasion: block IFN, hide from T cells, antigenic variation, latency.
- RT-PCR = gold standard (Ct, 2^ΔCt); RAT = fast but less sensitive; serology = past exposure.

⭐ Important questions

1. Explain the effects of viral infection on host cells and organisms.
2. Explain the interferon-induced antiviral state with a diagram.
3. Explain the counter-mechanisms used by viruses to evade the host's immune response.
4. Explain the principle and steps of RT-PCR for viral diagnosis.
5. Compare RT-PCR, rapid antigen test and serology.

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UNIT 3 Classification, Positive-Strand RNA and DNA Viruses

✍️ 2-mark definitions

- **ICTV:** the International Committee on Taxonomy of Viruses, which classifies and names viruses (realm, kingdom, phylum, class, order, family, genus, species; family names end in -viridae, genus in -virus).
- **Baltimore classification:** grouping of viruses into seven classes by genome type and the route used to make mRNA (David Baltimore, 1971).
- **Positive-sense (+) RNA:** genome RNA with the same sense as mRNA; can be translated directly and is infectious by itself.
- **Polyprotein:** a long protein translated from one ORF and cut by viral proteases into functional proteins (picornaviruses, flaviviruses, coronaviruses).
- **IRES (internal ribosome entry site):** an RNA structure allowing cap-independent translation (picornaviruses, HCV).
- **Arbovirus:** an arthropod-borne virus transmitted by mosquitoes, ticks or sandflies (dengue, Zika, JE, chikungunya).
- **ACE2:** angiotensin-converting enzyme 2, the human cell receptor used by SARS-CoV and SARS-CoV-2 spike proteins.
- **Episome:** a viral DNA genome maintained as an extra-chromosomal circle in the nucleus (HPV, EBV, HBV cccDNA).
- **Oncogenic virus:** a virus that can cause cancer (HPV → cervical cancer; HBV/HCV → liver cancer; EBV → Burkitt lymphoma).
- **Latency (herpesviruses):** the viral genome persists without virion production and can reactivate (HSV in sensory ganglia; VZV → shingles).

💡 Core concepts

Baltimore classification

Class	Genome	mRNA route	Examples
I	dsDNA	Host RNA pol II transcribes	Herpes, adeno, pox, papilloma
II	ssDNA	Converted to dsDNA, then transcribed	Parvovirus B19, circovirus
III	dsRNA	Viral RdRp transcribes	Rotavirus (reo)
IV	+ssRNA	Genome acts as mRNA	Polio, dengue, SARS-CoV-2, HCV
V	−ssRNA	Viral RdRp copies to + mRNA	Influenza, measles, rabies, Ebola
VI	ssRNA-RT	RNA → DNA by reverse transcriptase, integrates	HIV, HTLV
VII	dsDNA-RT	Replicates via RNA intermediate	Hepatitis B virus

Nomenclature rules (ICTV): order -virales, family -viridae, subfamily -virinae, genus -virus; species names now binomial (e.g. *Betacoronavirus pandemicum* is the SARS-CoV-2 species since 2023); vernacular names (virus names) are different from species names.

Picornaviridae ("pico" = small RNA)

- ~30 nm, non-enveloped icosahedral; +ssRNA ~7.5 kb with VPg (protein) at the 5′ end and poly-A at 3′.
- Replication in cytoplasm: IRES-driven translation of one polyprotein → cleaved by 2A, 3C proteases → RdRp (3D) makes – strand then + strands → assembly → lysis.
- Members: poliovirus (paralysis; OPV/IPV), rhinovirus (common cold), hepatitis A virus, coxsackie and enteroviruses (hand-foot-mouth disease), foot-and-mouth disease virus.

Flaviviridae

- ~50 nm, enveloped, icosahedral; +ssRNA ~11 kb, capped, single ORF → polyprotein (structural C, prM, E; non-structural NS1–NS5) cleaved by host and viral NS2B-NS3 protease; NS5 = RdRp and methyltransferase.
- Entry by receptor-mediated endocytosis and low-pH fusion via E protein; assembly in ER; maturation by furin cleavage of prM.
- Members: dengue (4 serotypes; antibody-dependent enhancement; Aedes aegypti), Zika (microcephaly), Japanese encephalitis (Culex; JE vaccine), yellow fever (17D live vaccine), West Nile; hepatitis C virus (genus Hepacivirus; chronic hepatitis; cured by direct-acting antivirals like sofosbuvir).

Coronaviridae (SARS-CoV-2)

- ~100–120 nm, enveloped, crown-like spikes; +ssRNA ~30 kb (among the largest RNA genomes), capped and poly-A.
- Structural proteins: **S** (spike: S1 binds ACE2 via RBD; S2 fusion; primed by furin and TMPRSS2), **E** (envelope), **M** (membrane), **N** (nucleocapsid).
- Replication: ORF1a/1ab translated into polyproteins pp1a/pp1ab → cleaved by 3CL^pro (Mpro) and PL^pro into 16 nsps → replication-transcription complex in double-membrane vesicles (nsp12 RdRp, nsp14 proofreading exonuclease) → nested subgenomic mRNAs for structural proteins → assembly in ERGIC → exocytosis.
- Diseases: SARS (2002–03), MERS (2012), COVID-19 (2019–); seasonal HCoVs cause colds. Variants (Alpha, Delta, Omicron) arose by spike mutations.

Small DNA viruses

Family	Genome	Features	Disease
Parvoviridae	ssDNA ~5 kb, linear	~20 nm, non-enveloped; needs dividing cells (S phase) or helper virus (AAV)	B19: fifth disease, aplastic crisis; AAV used as gene therapy vector
Papillomaviridae	dsDNA ~8 kb, circular	Non-enveloped; infects basal epithelium; E6 degrades p53, E7 inactivates Rb	Warts; HPV 16, 18 → cervical cancer (HPV vaccine)
Polyomaviridae	dsDNA ~5 kb, circular	Large T antigen binds p53 and Rb	JC virus (PML), BK virus (kidney), SV40

Large DNA viruses

Family	Genome	Features	Disease
Herpesviridae	dsDNA 125–240 kb, linear	Enveloped; replicate in nucleus; immediate-early, early, late genes; **latency**	HSV-1/2 (cold sores, genital herpes), VZV (chickenpox, shingles), CMV, EBV (mononucleosis, lymphoma), HHV-8 (Kaposi sarcoma)
Poxviridae	dsDNA 130–375 kb	Largest animal viruses; brick-shaped; replicate in **cytoplasm** with own RNA polymerase	Smallpox (eradicated 1980), mpox, vaccinia (vaccine)
Adenoviridae	dsDNA 26–45 kb	Non-enveloped icosahedral with fibres	Respiratory and eye infections; vectors for vaccines (ChAdOx1) and gene therapy

Miscellaneous agents: hepatitis B virus (Hepadnaviridae, partly dsDNA, replicates by reverse transcription, cccDNA in nucleus); hepatitis D (satellite needing HBV); viroids; prions; giant viruses (Mimivirus); bacteriophages (lytic and lysogenic cycles).

🧮 Solved examples

Example 1: Classify by Baltimore. A virus with a negative-sense, segmented RNA genome that carries its own RdRp → **class V** (e.g. influenza). A virus with dsDNA that replicates through an RNA intermediate → **class VII** (HBV).

Example 2: Genome coding capacity. An average protein of 300 amino acids needs 900 nucleotides. A 30 kb coronavirus genome could code roughly 30,000/900 ≈ **33 proteins** (it actually encodes about 29); a 7.5 kb picornavirus about 8 (it makes ~11 mature proteins from one polyprotein by cleavage).

Example 3: Dengue serotypes. A person infected first with DENV-2 later catches DENV-3. Non-neutralizing cross-reactive antibodies can help the virus enter Fc-receptor cells (**antibody-dependent enhancement**), raising the risk of severe dengue; this is why dengue vaccines must protect against all four serotypes.

Key points to remember

- Baltimore I–VII by genome and mRNA route; ICTV ranks and suffixes.
- Picorna: small, naked, +RNA, VPg, IRES, polyprotein (polio, rhino, HAV).
- Flavi: enveloped +RNA, polyprotein, NS3 protease, NS5 RdRp (dengue, Zika, JE, HCV).
- Corona: ~30 kb +RNA, spike-ACE2, TMPRSS2, Mpro, proofreading nsp14, subgenomic mRNAs.
- Small DNA: parvo (ssDNA), papilloma (E6/E7), polyoma; large DNA: herpes (latency), pox (cytoplasm), adeno (vectors).

⭐ Important questions

1. Explain the Baltimore classification and ICTV nomenclature of viruses.
2. Describe the structure and replication of picornaviruses (poliovirus).
3. Explain flaviviruses with dengue as an example.

NEXT ENGINEER _sheerat_

4. Describe the structure and replication of SARS-CoV-2.
5. Write notes on small DNA viruses (HPV) and large DNA viruses (herpes, pox).

NEXT ENGINEER
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UNIT 4 Negative-Strand RNA, dsRNA Viruses and Retroviruses

✍️ 2-mark definitions

- **Negative-sense (−) RNA virus:** a virus whose genome is complementary to mRNA; it must carry its own RNA-dependent RNA polymerase into the cell.
- **Segmented genome:** a genome divided into separate RNA molecules (influenza 8 segments; rotavirus 11 segments).
- **Antigenic drift:** small, gradual changes in influenza HA and NA due to point mutations, causing seasonal epidemics.
- **Antigenic shift:** a sudden major change in influenza A HA and/or NA subtype by reassortment of gene segments between strains (e.g. human and avian), which can cause pandemics.
- **Haemagglutinin (HA):** influenza surface glycoprotein that binds sialic acid receptors and mediates fusion.
- **Neuraminidase (NA):** influenza enzyme that cleaves sialic acid to release new virions; target of oseltamivir.
- **Retrovirus:** an RNA virus that uses reverse transcriptase to make a DNA copy of its genome, which integrates into the host chromosome as a provirus.
- **Reverse transcriptase (RT):** an RNA-dependent DNA polymerase (with RNase H activity) that converts viral RNA into dsDNA.
- **Provirus:** retroviral DNA integrated into the host genome.
- **AIDS:** acquired immunodeficiency syndrome, the late stage of HIV infection with $CD4^+$ T cells below 200 per µL or AIDS-defining illnesses.

💡 Core concepts

Common features of −ssRNA viruses: enveloped; helical nucleocapsid (RNA + N protein); virion carries RdRp (L protein + P); first event is transcription of mRNAs; later replication through a full-length + antigenome.

Paramyxoviridae

- Non-segmented −ssRNA (~15 kb); envelope with attachment protein (HN, H or G) and **F (fusion) protein**, which fuses at neutral pH at the cell surface → syncytia.
- Replication entirely in the cytoplasm; gene order 3′-N-P-M-F-HN-L-5′; transcription gradient (more N than L).
- Members: measles (MMR vaccine; Koplik spots; SSPE), mumps, parainfluenza, **Nipah virus** (henipavirus; bats → humans; outbreaks in Kerala since 2018; high fatality). (RSV is now in Pneumoviridae.)

Orthomyxoviridae (influenza)

- Enveloped, pleomorphic; 8 −ssRNA segments (influenza A and B) coding PB2, PB1, PA (polymerase), HA, NP, NA, M1/M2, NS1/NEP.

- Replication in the **nucleus**: entry by endocytosis, M2 ion channel acidifies the core, HA-mediated fusion, vRNPs enter nucleus; **cap-snatching** (PB2 binds host caps, PA cleaves them) primes viral mRNA synthesis; assembly at the plasma membrane; NA releases virions.
- Subtypes by HA (H1–H18) and NA (N1–N11); pandemics 1918 (H1N1), 1957 (H2N2), 1968 (H3N2), 2009 (H1N1pdm); avian H5N1, H7N9.
- Drift → annual vaccine updates; shift → pandemics. Drugs: oseltamivir, zanamivir (NA inhibitors), baloxavir (PA endonuclease inhibitor), amantadine (M2, now resistant).

Rhabdoviridae (rabies)

- Bullet-shaped, enveloped; −ssRNA ~12 kb with 5 genes N, P, M, G, L; G glycoprotein binds nicotinic acetylcholine receptor and others.
- Transmission by bite of infected animals (dogs cause 99% of human cases in India); travels by retrograde axonal transport to the CNS; Negri bodies in neurons; almost 100% fatal once symptoms appear.
- Prevention: wound washing, post-exposure prophylaxis (cell-culture rabies vaccine + rabies immunoglobulin for category III bites), pre-exposure vaccination; dog vaccination ("zero by 30" goal).

Reoviridae / rotavirus (dsRNA)

- Non-enveloped, triple-layered icosahedral capsid; 11 dsRNA segments.
- The dsRNA genome stays inside the inner core particle (hiding from PRRs); core RdRp transcribes capped mRNAs extruded through channels; replication in cytoplasmic viroplasms.
- NSP4 acts as an enterotoxin → secretory diarrhoea; leading cause of severe diarrhoea in infants.
- Vaccines: Rotavac (Indian, Bharat Biotech), Rotasiil (Serum Institute), RotaTeq, Rotarix; in India's Universal Immunization Programme.

Retroviruses and HIV

- Structure: enveloped ~100 nm; cone-shaped capsid (p24); diploid +ssRNA (~9.7 kb); enzymes RT, integrase, protease; envelope gp120 (SU) + gp41 (TM).
- Genes: **gag** (structural: MA, CA, NC), **pol** (protease, RT, integrase), **env** (gp160 → gp120 + gp41); accessory/regulatory: tat, rev, nef, vif, vpr, vpu.
- **Life cycle (draw it):** 1. gp120 binds CD4 and co-receptor CCR5 or CXCR4. 2. gp41 mediates membrane fusion; capsid enters. 3. Reverse transcription: RNA → ssDNA → dsDNA (error-prone → high mutation rate). 4. Nuclear import and integration by integrase → provirus. 5. Transcription by host RNA pol II (Tat boosts; Rev exports unspliced RNA). 6. Translation of Gag and Gag-Pol polyproteins; Env processed in ER/Golgi. 7. Assembly at plasma membrane and budding. 8. Maturation: viral protease cleaves Gag and Gag-Pol → infectious virion.
- **HIV/AIDS:** transmission by sex, blood, mother-to-child; acute phase (flu-like, high viraemia) → clinical latency (years, CD4 decline) → AIDS (CD4 < 200, opportunistic infections: TB, Pneumocystis, candidiasis, Kaposi sarcoma).
- **Treatment (ART):** combination of NRTIs (tenofovir, lamivudine, zidovudine), NNRTIs (efavirenz, nevirapine), protease inhibitors (lopinavir, darunavir), integrase inhibitors (dolutegravir), entry inhibitors (maraviroc, enfuvirtide). India's first-line: TLD (tenofovir + lamivudine + dolutegravir). "Undetectable = untransmittable".
- Diagnosis: ELISA/rapid antibody tests, Western blot (older), p24 antigen, viral load by RT-PCR, CD4 count.

🧮 Solved examples

Example 1: Influenza reassortment. A pig cell co-infected with human H3N2 and avian H5N1 can produce progeny with any combination of the 8 segments: 2^8 = **256** possible genotypes, including new HA/NA combinations like H5N2 or H3N1 (antigenic shift).

Example 2: HIV mutation rate. HIV RT makes about 3×10^{-5} errors per nucleotide per cycle; with a 9,700-nt genome, each new genome carries about $9{,}700 \times 3 \times 10^{-5} \approx$ **0.3 mutations**, so about one in three progeny carries a new mutation. With billions of virions a day, every single drug-resistance mutation arises daily → need for **combination therapy**.

Example 3: CD4 count. A patient has 1,800 lymphocytes/µL with 10% $CD4^+$ cells → CD4 = **180/µL**, below 200 → AIDS-defining by count; needs ART and prophylaxis against opportunistic infections.

Key points to remember

- −RNA viruses carry RdRp; transcription first.
- Paramyxo: F protein, syncytia, cytoplasmic; measles, mumps, Nipah.
- Influenza: 8 segments, nuclear replication, cap-snatching, HA/NA, drift vs shift, NA inhibitors.
- Rabies: bullet shape, N-P-M-G-L, retrograde axonal transport, Negri bodies, PEP.
- Rotavirus: 11 dsRNA segments, triple capsid, NSP4 enterotoxin, Rotavac.
- HIV: gag-pol-env, CD4 + CCR5/CXCR4, RT, integrase, protease; ART combination.

⭐ Important questions

1. Explain the general features of negative-strand RNA viruses. Describe paramyxoviruses.
2. Describe the structure and replication of influenza virus. Explain antigenic drift and shift.
3. Explain rabies virus: structure, pathogenesis and prevention.
4. Explain rotavirus as a dsRNA virus.
5. Describe the structure and life cycle of HIV and the pathogenesis of AIDS.

UNIT 5 Vaccines, Antivirals and RNA-Based Therapies

✍️ 2-mark definitions

- **Vaccine:** a biological preparation that trains the immune system to recognize a pathogen and protect against future infection, without causing the disease.
- **Live attenuated vaccine:** a weakened live virus that replicates without causing disease (OPV, MMR, yellow fever 17D, rotavirus, varicella).
- **Inactivated (killed) vaccine:** virus killed by heat or chemicals (formaldehyde, β-propiolactone) (IPV, rabies cell-culture vaccine, Covaxin).
- **Subunit vaccine:** contains only purified or recombinant viral proteins (hepatitis B HBsAg vaccine; HPV virus-like particles).
- **mRNA vaccine:** lipid-nanoparticle-delivered mRNA encoding a viral antigen that host cells translate (COVID-19 vaccines by Pfizer-BioNTech and Moderna).
- **Adjuvant:** a substance added to a vaccine to strengthen and direct the immune response (alum, MF59, AS01, AS04, CpG).
- **Antiviral drug:** a drug that inhibits a specific step of the virus life cycle with acceptable toxicity to host cells.
- **Antisense oligonucleotide (ASO):** a short single-stranded DNA/RNA analogue complementary to a target mRNA that blocks its translation or triggers its degradation (RNase H).
- **siRNA (small interfering RNA):** a 21–23 nt double-stranded RNA that guides the RISC complex (Argonaute) to cleave a complementary mRNA.
- **Ribozyme:** a catalytic RNA molecule that cleaves specific RNA sequences (hammerhead, hairpin ribozymes).

💡 Core concepts

Types of viral vaccines

Type	Examples	Pros	Cons
Live attenuated	OPV, MMR, yellow fever, varicella, rotavirus, nasal influenza	Strong, long-lasting cellular and humoral immunity; often one dose; cheap	Reversion risk (vaccine-derived polio); not for immunocompromised or pregnant; cold chain
Inactivated	IPV, rabies, hepatitis A, Covaxin, injectable flu	Safe, no reversion	Weaker, needs boosters and adjuvants
Subunit / recombinant protein	Hepatitis B, Novavax, Corbevax (India)	Very safe, defined	Needs adjuvant, boosters

Type	Examples	Pros	Cons
Virus-like particles (VLPs)	HPV (Gardasil, Cervavac — India's indigenous HPV vaccine)	Highly immunogenic, non-infectious	Complex production
Viral vector	Covishield (ChAdOx1), Sputnik V, Ebola rVSV-ZEBOV	Strong T-cell response	Pre-existing immunity to vector
mRNA	Pfizer-BioNTech, Moderna, India's Gemcovac	Rapid design and manufacture	Cold chain (lipid nanoparticles), cost
DNA	ZyCoV-D (India, 2021: world's first DNA vaccine approved for humans)	Stable, easy to make	Lower immunogenicity, needle-free injector

Adjuvants: aluminium salts (alum: depot effect, inflammasome activation); oil-in-water emulsions (MF59, AS03); AS01 (MPL + QS-21 in liposomes; shingles and malaria vaccines); AS04 (MPL + alum; HPV); TLR agonists (CpG 1018 in hepatitis B vaccine; Algel-IMDG in Covaxin). Functions: enhance antigen uptake, activate innate immunity, prolong antigen exposure, reduce dose, direct Th1/Th2 response.

Vaccine manufacturing

1. Master and working seed lots (virus) and cell banks (Vero, MRC-5) or eggs.
2. Upstream: virus propagation in bioreactors or eggs, or recombinant protein expression (yeast, insect, CHO cells), or mRNA in vitro transcription.
3. Downstream: harvest, clarification, purification (chromatography, ultrafiltration), inactivation (if inactivated vaccine).
4. Formulation with adjuvant, stabilizers, preservatives; fill-finish in vials; lyophilization if needed.
5. Packaging, cold chain (2–8 °C; −20/−70 °C for some mRNA vaccines).

Quality assurance and quality control (QA/QC): Good Manufacturing Practice (GMP); tests for identity, **potency** (antigen content, animal or cell-based assays), **sterility**, purity (host-cell protein and DNA limits), **safety** (abnormal toxicity, absence of adventitious agents, complete inactivation tests), endotoxin (LAL test), pyrogens, stability and shelf-life, adjuvant content, pH, appearance. Lot release by the national control laboratory (Central Drugs Laboratory, Kasauli) after CDSCO approval; WHO prequalification for global supply; vaccine vial monitors; post-marketing surveillance (AEFI).

Antivirals: targets in the life cycle

Step	Drug (virus)
Attachment/entry	Maraviroc (HIV CCR5), enfuvirtide (HIV gp41 fusion), palivizumab and nirsevimab (RSV antibodies)
Uncoating	Amantadine (influenza M2; resistance now common)

Step	Drug (virus)
Genome replication: nucleoside/nucleotide analogues	Acyclovir (HSV/VZV; activated by viral thymidine kinase → selective), ganciclovir (CMV), zidovudine, lamivudine, tenofovir (HIV/HBV), sofosbuvir (HCV NS5B), remdesivir (SARS-CoV-2 RdRp), molnupiravir (lethal mutagenesis), favipiravir
Non-nucleoside RT inhibitors	Nevirapine, efavirenz (HIV)
Integration	Raltegravir, dolutegravir (HIV)
Cap-snatching	Baloxavir (influenza PA)
Polyprotein processing	Lopinavir, darunavir (HIV protease); nirmatrelvir (SARS-CoV-2 Mpro, with ritonavir as Paxlovid); glecaprevir (HCV NS3/4A)
Release	Oseltamivir, zanamivir (influenza neuraminidase)
Immune modulation	Interferon-α (HBV, HCV earlier), pegylated interferon

Interferons as drugs: recombinant IFN-α (and pegylated IFN-α with longer half-life) for chronic hepatitis B and earlier HCV; IFN-β for multiple sclerosis; side effects: flu-like symptoms, depression, bone marrow suppression.

Screening of antivirals: target-based (enzyme assays on RT, protease, polymerase) and cell-based (CPE inhibition, plaque reduction, reporter viruses) high-throughput screening; measure **EC_{50}** (effective concentration) and **CC_{50}** (cytotoxic concentration); **selectivity index SI = CC_{50}/EC_{50}** (higher is safer, > 10 desired); then animal models and clinical trials (phases I–III).

Antiviral resistance: arises from high mutation rates and selection under drug pressure: examples: HIV M184V (lamivudine), K103N (NNRTIs); influenza H275Y (oseltamivir), S31N (amantadine); HSV thymidine kinase mutants (acyclovir). Prevention: combination therapy, adherence, high-barrier drugs, resistance testing.

Antisense, siRNA and ribozymes

Approach	Mechanism	Example
Antisense oligonucleotides	Bind target mRNA by base pairing → block translation or RNase H cleavage; chemically modified (phosphorothioate, 2′-O-methyl) for stability	Fomivirsen (CMV retinitis, first approved antisense drug, 1998)
siRNA (RNA interference)	dsRNA loaded into RISC; guide strand directs Argonaute-2 to cleave complementary viral mRNA	Approved for genetic diseases (patisiran); experimental against HBV, RSV, SARS-CoV-2
Ribozymes	Catalytic RNA that base-pairs with and cleaves target RNA (hammerhead, hairpin)	Tested in HIV gene therapy

Challenges: delivery into cells (lipid nanoparticles, GalNAc conjugates for liver), stability against nucleases, off-target effects, innate immune activation, viral escape mutations (use several targets).

🧮 Solved examples

Example 1: Herd immunity threshold. For R_0 = 3, the threshold = 1 − 1/R_0 = **66.7%** of the population must be immune. With a vaccine of 90% efficacy, coverage needed = 0.667/0.9 = **74%**. For measles (R_0 ≈ 15): 1 − 1/15 = 93.3% immune; with 95% efficacy, coverage ≈ 98%.

Example 2: Selectivity index. A compound has EC_{50} = 0.5 µM and CC_{50} = 200 µM. SI = 200/0.5 = **400** → highly selective, a good lead. Another with EC_{50} = 10 µM, CC_{50} = 50 µM has SI = 5 → too toxic.

Example 3: Vaccine efficacy. In a trial, 100 of 10,000 placebo recipients and 10 of 10,000 vaccinees get the disease. Efficacy = (1 − 0.001/0.01) × 100 = **90%**.

Key points to remember

- Vaccine types: live, inactivated, subunit, VLP, vector, mRNA, DNA (Indian examples: Covaxin, Covishield, ZyCoV-D, Corbevax, Cervavac, Rotavac).
- Adjuvants: alum, MF59, AS01, AS04, CpG.
- Manufacturing: seed lot → upstream → downstream → formulation → fill-finish; QC: identity, potency, sterility, purity, safety, endotoxin, stability; CDL Kasauli lot release.
- Antivirals by life-cycle step; acyclovir selective via viral TK; oseltamivir NA; remdesivir RdRp.
- SI = CC_{50}/EC_{50}; resistance → combination therapy.
- Antisense (fomivirsen), siRNA (RISC), ribozymes (catalytic RNA).

⭐ Important questions

1. Classify viral vaccines with examples, advantages and limitations.
2. Explain the role of adjuvants and modern vaccine technologies (mRNA, vector, DNA).
3. Explain vaccine manufacturing and QA/QC.
4. Explain the mechanisms of antiviral drugs, interferons, antiviral screening and resistance.
5. Explain antisense RNA, siRNA and ribozymes as antiviral strategies.

Last-hour checklist

☐ History table (Ivanovsky, Beijerinck, Stanley, Jenner, Pasteur); properties; symmetry; sizes; genome types.

☐ Isolation (cells, eggs, animals); purification (centrifugation, gradients); plaque titre (4.5×10^8 PFU/mL); MOI (5 → 99.3%).

☐ Effects on host (CPE, latency, transformation); IFN pathway; evasion table.

☐ RT-PCR steps and Ct (32-fold); RAT; serology; PPV 67.8%, NPV 98.9%.

☐ Baltimore I–VII; ICTV naming; picorna, flavi, corona (spike-ACE2, TMPRSS2, Mpro); parvo, papilloma (E6/E7), polyoma; herpes (latency), pox (cytoplasm), adeno.

☐ Paramyxo (F protein, Nipah); influenza (8 segments, drift/shift, 256 reassortants); rabies (N-P-M-G-L, PEP); rotavirus (11 dsRNA, NSP4); HIV life cycle and ART; CD4 180.

☐ Vaccine types table with Indian examples; adjuvants; manufacturing and QC; herd immunity (66.7%, 74%); efficacy 90%.

☐ Antiviral targets table; IFN therapy; SI 400; resistance mutations; antisense, siRNA, ribozymes.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Antiviral Agents, Vaccines, and Immunotherapies	Stephen K. Tyring	CRC Press (Marcel Dekker)	WorldCat
2	Basic Virology	Edward K. Wagner, Martinez J. Hewlett	Blackwell Science	WorldCat
3	Fundamentals of Molecular Virology	Nicholas H. Acheson	Wiley (2011)	WorldCat
4	Principles of Virology, 2nd ed.	S. J. Flint, L. W. Enquist, R. M. Krug, V. R. Racaniello, A. M. Skalka	ASM Press	WorldCat
5	Medical Virology, 4th ed.	David O. White, Frank J. Fenner	Academic Press	WorldCat

Syllabus source: AKTU Open Elective-III and IV syllabus for B.Tech 8th semester, 2025-26 (copy hosted by an affiliated college, PDF)